Minimal Cores and Pareto Frontiers of Exact Pushdown Lifts

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Abstract

Exact pushdown realizations of the same semantic transition system need not admit a meaningful representation-independent notion of minimality: redundant coordinates can be added cheaply, while sufficiently broad behavioral quotients erase the pushdown structure one wants to compare. We introduce a protected comparison theory for finite-coordinate exact sectioned pushdown lifts. Strict coordinate quotients (SCQ) merge finite-control and stack-symbol coordinates letterwise while preserving the unique permanent bottom marker, semantics, transitions, and the chosen section; every strict quotient decreases the used state/symbol resource pair and every finite lift has a minimal quotient core. Bounded bi-recodings (BBR) are treated separately as invertible physical equivalences, leading to finite recoding-invariant Pareto envelopes. A finite protected interface characterizes the maximal configuration quotients that do not erase declared physical information, and protected representation contracts order both admissible architectures and observables.

We prove Pareto monotonicity under contract refinement and a strictness theorem based on resource down-set exclusion; protected-role capacity bounds provide quantitative certificates. For rational balance semantics, the framework yields an exact state premium between broad and deferred literal-demand contracts and exact two-resource frontiers for raw residual-faithful and normalized phase-local architectures. The raw Model-R frontier is $\{(p+q, 2), (\max(p, q), 3)\}$, while each normalized phase-local contract has an explicit staircase Pareto antichain. We identify precisely when extra stack symbols absorb the state cost of stronger local phase visibility. The results separate quotient minimality, harmless recoding, protected information, and unrestricted recognition complexity.

Keywords. deterministic pushdown automata; exact lifts; minimization; representation contracts; stack recoding; Pareto frontiers; descriptional complexity.

1 Introduction

The phrase "minimal pushdown representation" is dangerous unless the comparison category is stated. If two exact lifts of the same semantic system are compared only up to isomorphism, cheap redundant coordinates create infinitely many irrelevant variants. If arbitrary behavioral quotients are allowed, the comparison can collapse directly to semantics and erase the LIFO structure that was meant to be measured.

Deterministic pushdown equivalence is decidable in the classical sense [3], and restricted pushdown models such as visibly pushdown automata have their own minimization theory [4, 5]; neither question by itself fixes which physical state/stack information an exact normalization lift is required to preserve. Standard pushdown terminology is as in [2]. This paper develops a middle theory. Genuine reductions are coordinate quotients that preserve a chosen exact section and a declared physical interface. Harmless bounded stack recodings are not reductions at all; they are treated as equivalences. The resulting resource object is therefore not necessarily one pair $(|Q|, |\Gamma|)$ but a finite Pareto antichain associated with a recoding class.

The framework is motivated by the rational $p : q$ normalization program, where the same residual semantics admits literal-demand, raw-residual, phase-local, eager, and lazy realizations with different resource profiles. The point is not to select one representation as universally minimal. The point is to make the preserved information explicit and then prove what minimality means inside that contract.

2 From the original 2:1 machine to representation minimality

The need for a category-relative notion of minimality is already visible in the author's original completion-aware 2:1 DPDA, shown in fig. 1. The machine stores one semantic deficit jointly in finite control and stack content: a side state can carry a bounded correction even when the visible logical stack is small, and the end-of-input completion rules transfer that correction back into the store.

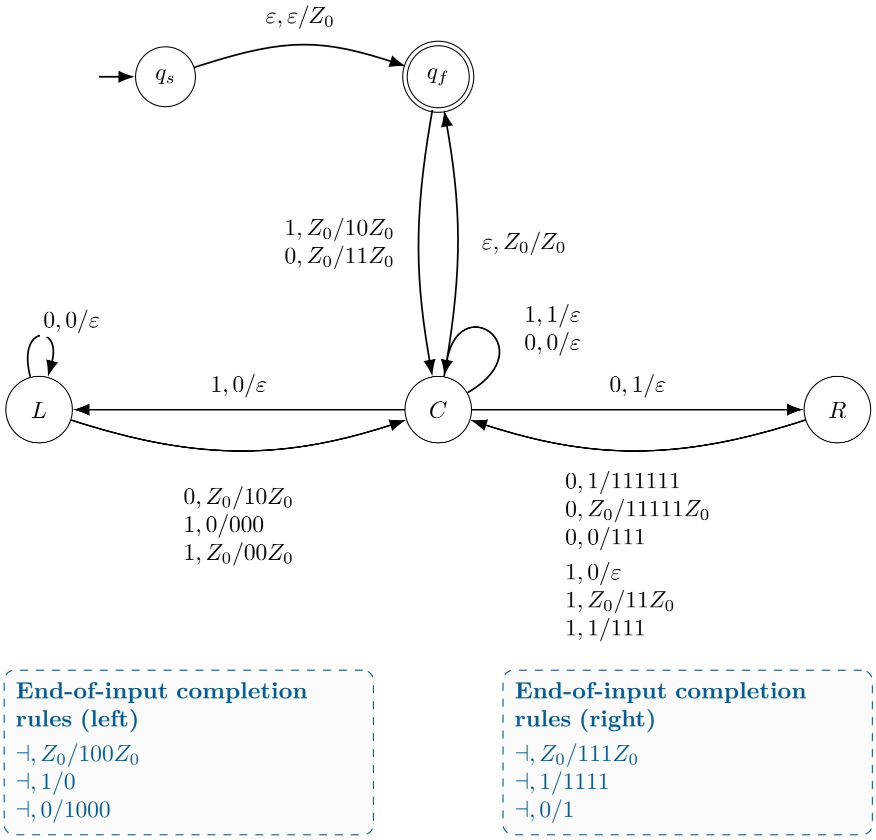

Figure 1: The original completion-aware 2:1 DPDA, in the empty-word-included formal variant. The same semantic deficit is distributed between control state and stack content, foreshadowing the representation-dependence that motivates protected quotient and Pareto theory.

The subsequent $p{:}q$ program made the representation issue unavoidable. The same residual semantics can be realized by literal-demand, eager or lazy, raw residual-faithful, and normalized phase-local architectures, and the resource counts change when information is moved between control, visible stack symbols, hidden stack structure, and normalization. Hence the question "which DPDA is minimal?" has no useful answer until one says which physical information must be preserved and which recodings are regarded as harmless.

The present framework is the formal endpoint of that progression:

$$\boxed{\begin{array}{c}\text{original 2:1 split state/stack encoding} \longrightarrow \text{multiple exact } p{:}q \text{ representations} \\ \longrightarrow \text{category-relative resource claims} \longrightarrow \text{protected minimal cores and Pareto frontiers}\end{array}}$$

The original transition table is not used in the abstract quotient theorems. Its role is to exhibit the concrete phenomenon that motivated the distinction between semantic equivalence, physical representation, quotient reduction, and recoding equivalence.

3 Exact sectioned lifts and used coordinates

Let M be a semantic transition system over input alphabet Σ. An exact sectioned pushdown lift is a tuple

$$\mathcal{L} = (C^{\mathrm{reach}}, \delta, \Theta, \sigma),$$

where C^{reach} is the reachable physical configuration set, δ is the one-top pushdown transition structure, $\Theta : C^{\mathrm{reach}} \to M$ is the semantic projection, and $\sigma : M \to C^{\mathrm{reach}}$ is a chosen section satisfying

$$\Theta\sigma = \mathrm{id}_M.$$

Exactness means that the physical transitions project to the semantic transitions.

For a finite-coordinate lift, define

$$Q_{\mathrm{use}} = \{q : \exists S\ (q, S) \in C^{\mathrm{reach}}\}$$

and let Γ_{use} be the stack symbols appearing in reachable stacks, including exactly one permanent bottom marker Z_0.

The basic resource vector is

$$\mathbf{R}(\mathcal{L}) = (|Q_{\mathrm{use}}|, |\Gamma_{\mathrm{use}}|).$$

Unreachable ambient states and symbols are ignored.

4 Strict coordinate quotients

Definition 4.1 (Strict coordinate quotient). *Let $\mathcal{L}_1, \mathcal{L}_2$ be exact sectioned one-top lifts of the same semantic system. A strict coordinate quotient (SCQ)*

$$F : \mathcal{L}_1 \twoheadrightarrow_{\mathrm{SCQ}} \mathcal{L}_2$$

consists of surjections

$$f_Q : Q^{(1)}_{\mathrm{use}} \twoheadrightarrow Q^{(2)}_{\mathrm{use}}, \qquad f_\Gamma : \Gamma^{(1)}_{\mathrm{use}} \twoheadrightarrow \Gamma^{(2)}_{\mathrm{use}}$$

such that

$$f_\Gamma^{-1}(Z_0) = \{Z_0\},$$

and such that the induced stack map is letterwise and non-erasing:

$$F(q, X_1 \cdots X_k Z_0) = \big(f_Q(q), f_\Gamma(X_1) \cdots f_\Gamma(X_k) Z_0\big).$$

It must satisfy

$$F(C_1^{\mathrm{reach}}) = C_2^{\mathrm{reach}}, \quad \Theta_2 F = \Theta_1, \quad F\sigma_1 = \sigma_2,$$

and

$$F\delta_a^{(1)} = \delta_a^{(2)} F$$

on every reachable transition.

The bottom-marker condition is essential: a work symbol may not be merged with the unique permanent bottom cell.

Theorem 4.2 (SCQ reduction theorem)**.** *SCQ morphisms compose and define a preorder. Mutual SCQ reducibility is exactly SCQ isomorphism. If*

$$\mathcal{L}_1 \twoheadrightarrow_{\mathrm{SCQ}} \mathcal{L}_2,$$

then

$$\mathbf{R}(\mathcal{L}_2) \leq \mathbf{R}(\mathcal{L}_1)$$

componentwise; if the quotient is not an isomorphism, at least one coordinate decreases strictly.

Proof. Composition is coordinatewise and preserves the transition, semantic, section, and bottom-marker conditions. If SCQ morphisms exist in both directions, finiteness and surjectivity force equality of the used state and symbol cardinalities; all coordinate maps are bijective and their inverses preserve the same structure. For monotonicity, each coordinate map is surjective. Equality in both resource coordinates forces both maps to be bijections, hence an isomorphism. □

Corollary 4.3 (Finite minimal-core existence)**.** *Every finite-coordinate exact sectioned lift has at least one SCQ-minimal quotient descendant.*

Proof. Each strict SCQ step strictly decreases a point of $\mathbb{N}^2$. No infinite strictly descending chain exists. □

Cheap finite-control decorations are automatically removed: if a redundant cyclic coordinate $j \in \mathbb{Z}_m$ is added to control but ignored by semantics and stack dynamics, projection $(q, j) \mapsto q$ is a strict SCQ quotient. On the other hand, an arbitrary behavioral quotient that erases the LIFO store is excluded by the letterwise non-erasing coordinate requirement.

5 Protected interfaces and maximal safe quotients

SCQ is intentionally rigid. To formulate which physical distinctions a quotient is allowed to erase, we attach a finite interface signature.

Definition 5.1 (Protected-interface lift)**.** *A protected-interface lift is a pair*

$$(\mathcal{L}, \lambda), \qquad \lambda : C^{\mathrm{reach}} \to \Lambda,$$

where Λ is finite and λ records physical information declared to be part of the comparison contract.

For a literal one-top policy, a useful interface is

$$\lambda_{\mathrm{lit}}(c) = \big(o_{\mathrm{lit}}(c), \chi_\sigma(c)\big),$$

where o_{lit} is the locally visible state/top observation and

$$\chi_\sigma(c) = \mathbf{1}_{\{c \in \sigma(M)\}}$$

records whether the configuration lies on the chosen canonical section.

Theorem 5.2 (Maximal interface-safe quotient criterion)**.** *Let $F : C_1^{\mathrm{reach}} \twoheadrightarrow C_2^{\mathrm{reach}}$ be surjective and let $\lambda_1 : C_1^{\mathrm{reach}} \to \Lambda$. The following are equivalent:*

1. *there is a unique $\lambda_2 : C_2^{\mathrm{reach}} \to \Lambda$ such that*

$$\lambda_1 = \lambda_2 \circ F;$$

2. *λ_1 is constant on every fiber of F.*

Hence λ-separated configurations can never be identified by an interface-safe quotient.

Proof. Necessity is immediate. Conversely define $\lambda_2(F(c)) = \lambda_1(c)$; fiber constancy makes the definition well posed, and surjectivity gives uniqueness. □

This criterion is maximal at the configuration level: it imposes exactly the identifications forbidden by the protected observable and no others.

6 Bounded bi-recoding and recoding-invariant resource spectra

A bounded invertible change of stack code should not be counted as a reduction.

Definition 6.1 (Bounded bi-recoding)**.** *A bounded bi-recoding (BBR) between protected-interface lifts consists of a bijective control recoding and a non-erasing work-stack block code*

$$\varphi : \Gamma_{1,\mathrm{use}} \setminus \{Z_0\} \to (\Gamma_{2,\mathrm{use}} \setminus \{Z_0\})^+$$

extended multiplicatively above the fixed bottom marker, such that:

1. $1 \le |\varphi(X)| \le B$ *for a uniform finite* B*;*
2. *an inverse bounded non-erasing block code exists on the reachable stack languages;*
3. *the induced configuration bijection intertwines semantics, section, transitions, and the protected interface.*

BBR is therefore an actual conjugacy of reachable dynamics. It preserves every obstruction expressible in semantic fibers, paths, section membership, and protected-interface values.

The identity code, the stipulated bounded inverse, and composition of bounded block codes show that BBR is an equivalence relation; see Appendix A.

Define the protected recoding spectrum

$$\mathscr{R}_\lambda(\mathcal{L}) = \{(|Q_{\mathrm{use}}(\mathcal{L}')|, |\Gamma_{\mathrm{use}}(\mathcal{L}')|) : \mathcal{L}' \simeq_{\mathrm{BBR},\lambda} \mathcal{L}\}.$$

Its minimal antichain is

$$\mathcal{P}_\lambda(\mathcal{L}) = \mathrm{Min}\, \mathscr{R}_\lambda(\mathcal{L}).$$

Theorem 6.2 (Finite protected Pareto envelope)**.** *For every finite-coordinate protected lift,* $\mathcal{P}_\lambda(\mathcal{L})$ *is nonempty and finite and depends only on the BBR-equivalence class.*

Proof. The original presentation belongs to its spectrum, so the set is nonempty. Every nonempty subset of $\mathbb{N}^2$ has minimal elements, and Dickson's lemma [1] implies that the antichain of all componentwise-minimal elements is finite. BBR-equivalent lifts have the same spectrum by transitivity. □

A normalized protected core step first chooses a representative whose resource vector lies in $\mathcal{P}_\lambda$, then takes a strict λ-preserving SCQ quotient, and finally passes to the target BBR class. By Theorem 4.2, every such step strictly decreases the normalized resource pair. Hence terminal protected BBR-classes exist. Uniqueness is not asserted.

7 Protected representation contracts and refinement

Observable refinement alone cannot compare genuinely different physical re-coordinations. The representation class must also be part of the data.

Definition 7.1 (Observable refinement)**.** *On one representation class, write*

$$\lambda' \succeq_{\mathrm{obs}} \lambda$$

if there is a map h *with*

$$\lambda = h \circ \lambda'.$$

Then λ' *remembers at least as much protected information as* λ*.*

Proposition 7.2 (Quotient antitonicity)**.** *If* $\lambda' \succeq_{\mathrm{obs}} \lambda$*, every* λ'*-safe surjective quotient is* λ*-safe.*

Proof. If $\lambda' = \bar{\lambda}' F$, then $\lambda = h\bar{\lambda}' F$. □

Definition 7.3 (Protected representation contract)**.** *A protected contract is*

$$\mathfrak{K} = (\mathfrak{C}, \{\lambda_{\mathcal{L}}\}_{\mathcal{L}\in\mathfrak{C}}),$$

where $\mathfrak{C}$ is a BBR-closed class of finite-coordinate exact sectioned pushdown lifts of one semantic system and each $\lambda_{\mathcal{L}} : C^{\mathrm{reach}}_{\mathcal{L}} \to \Lambda$ is compatible with BBR conjugacy. We suppress the object subscript when harmless. Write $\mathfrak{K}' \succeq \mathfrak{K}$ if $\mathfrak{C}' \subseteq \mathfrak{C}$ and the protected interfaces refine objectwise on every lift of $\mathfrak{C}'$. Appendix A records the typing explicitly.

Define

$$\mathscr{R}(\mathfrak{K}) = \{(|Q_{\mathrm{use}}(\mathcal{L})|, |\Gamma_{\mathrm{use}}(\mathcal{L})|) : \mathcal{L} \in \mathfrak{C}\}$$

and

$$\mathrm{Pareto}(\mathfrak{K}) = \mathrm{Min}\,\mathscr{R}(\mathfrak{K}).$$

For finite antichains $A, B \subset \mathbb{N}^2$, write $A \preceq_P B$ if every $b \in B$ is componentwise dominated by some $a \in A$.

Theorem 7.4 (Contract-refinement Pareto monotonicity)**.** *If $\mathfrak{K}' \succeq \mathfrak{K}$, then*

$$\mathscr{R}(\mathfrak{K}') \subseteq \mathscr{R}(\mathfrak{K})$$

and

$$\boxed{\mathrm{Pareto}(\mathfrak{K}) \preceq_P \mathrm{Pareto}(\mathfrak{K}').}$$

Thus a stronger physical contract cannot improve the Pareto frontier.

Proof. Class inclusion gives resource-set inclusion. For any finer-contract Pareto point b, the same point belongs to the coarser resource set. Descending componentwise inside $\mathbb{N}^2$ yields a coarser minimal point $a \le b$. □

8 Strictness from protected-information capacity

Weak monotonicity is automatic from class inclusion. Strictness requires an actual obstruction.

Definition 8.1 (Resource down-set)**.** *For $r \in \mathbb{N}^2$, let*

$$\downarrow r = \{s \in \mathbb{N}^2 : s \le r \ \textit{componentwise}\}.$$

Theorem 8.2 (Strictness by down-set exclusion)**.** *Let $\mathfrak{K}' \succeq \mathfrak{K}$. If there exists an attainable coarse point $r_0 \in \mathscr{R}(\mathfrak{K})$ such that*

$$\mathscr{R}(\mathfrak{K}') \cap \downarrow r_0 = \varnothing,$$

then

$$\boxed{\mathrm{Pareto}(\mathfrak{K}) \prec_P \mathrm{Pareto}(\mathfrak{K}').}$$

Proof. Theorem 7.4 gives weak dominance. Since r_0 is coarse-attainable, some coarse Pareto point a satisfies $a \le r_0$. Reverse dominance would give a fine Pareto point $b \le a \le r_0$, contradicting the down-set exclusion. □

Corollary 8.3 (Protected-role capacity certificate)**.** *Suppose a fine contract forces N protected roles to occupy distinct local-observation cells, while every realization with resource vector at most r has at most* $\mathrm{cap}(r)$ *such cells, with* cap *monotone. If a coarse attainable point r_0 satisfies*

$$N > \mathrm{cap}(r_0),$$

then r_0 is a down-set exclusion witness and Pareto refinement is strict.

This is the main conceptual bridge:

$$\boxed{\text{protected-information separation} \Longrightarrow \text{resource down-set exclusion} \Longrightarrow \text{strict Pareto penalty.}}$$

9 Rational balance application I: an exact contract state premium

The abstract SCQ/BBR/protected-contract results above are proved entirely within this paper. The rational-balance applications now combine that new comparison framework with four exact endpoint resource theorems from the preceding normalization theory. Appendix B states those inputs and records their proof interfaces and scope; in particular, the unrestricted state lower bound and the raw/phase-local optima are not being re-labelled as new results of the present paper.

Fix coprime $p, q \geq 1$ and a mandatory right-endmarker convention. Let $\mathfrak{K}_{\text{broad}}$ be the broad exact-pushdown contract for the residual semantics of $L_{p,q}$. The previously established broad-class state theorem gives

$$Q_{\min}(\mathfrak{K}_{\text{broad}}) = \max(p, q).$$

A three-symbol exact realization attains this value.

Let $\mathfrak{K}_{\text{def}}$ be the canonical deferred-SEARCH literal-demand contract. Its previously established exact state minimum is

$$D_{p,q} = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1,$$

also attained with three total stack symbols.

Theorem 9.1 (Exact broad/deferred state premium)**.** *The exact state premium of the deferred literal-demand contract over the broad exact-pushdown contract is*

$$\boxed{\Delta_Q(p,q) = D_{p,q} - \max(p,q) = \min(p,q) - \mathbf{1}_{\{p+q \ even\}}.}$$

It is strict except when $\min(p, q) = 1$ *and* $p + q$ *is even.*

Proof. If $p + q$ is odd, $D_{p,q} = p + q$, giving $\Delta_Q = \min(p, q)$. If $p + q$ is even, $D_{p,q} = p + q - 1$, giving $\Delta_Q = \min(p, q) - 1$. The equality classification follows immediately. □

The two attaining constructions use the same total stack alphabet size 3, so the displayed witness points isolate a control-state premium rather than a stack-alphabet artifact.

10 Rational balance application II: exact two-resource frontiers

Let

$$M = \max(p, q).$$

In raw residual-faithful Model R, let C be the number of work symbols; the total stack alphabet has size $C + 1$ including Z_0. The exact per-C state minimum is

$$S_{\text{raw}}(C) = \begin{cases} p+q, & C = 1, \\ M, & C \geq 2. \end{cases}$$

Theorem 10.1 (Exact raw two-resource frontier)**.** *Across all finite work-alphabet sizes,*

$$\boxed{\text{Pareto}(\mathfrak{K}_{\text{raw}}) = \{(p+q, 2), (M, 3)\}.}$$

Proof. The point $C = 1$ gives $(p + q, 2)$. The point $C = 2$ gives $(M, 3)$ and dominates every $C > 2$ point $(M, C + 1)$. Since $M < p + q$, the two displayed points are incomparable. □

Now fix $B \geq M$ and write every residual as $r = s + Bz$, $0 \leq s < B$. Define

$$h_+(B) = \frac{p}{\gcd(B,p)}, \qquad h_-(B) = \frac{q}{\gcd(B,q)}, \qquad H(B) = h_+(B) + h_-(B).$$

The normalized phase-local contract requires the current signed quotient phase to be visible through control state plus one work symbol. Its exact per-C state minimum is

$$Q^{\min}_{\text{phase}}(B,C) = B\left\lceil \frac{H(B)}{C} \right\rceil.$$

Theorem 10.2 (Exact phase-local Pareto staircase)**.** *Let $H = H(B)$ and*

$$\mathcal{B}_H = \{1\} \cup \left\{2 \leq C \leq H : \left\lceil \frac{H}{C} \right\rceil < \left\lceil \frac{H}{C-1} \right\rceil \right\}.$$

Then

$$\boxed{\operatorname{Pareto}(\mathfrak{K}_{\text{phase}}(B)) = \left\{ \left(B\left\lceil \frac{H}{C} \right\rceil, C+1 \right) : C \in \mathcal{B}_H \right\}.}$$

Proof. The state coordinate is nonincreasing in C, while the alphabet coordinate is strictly increasing. A point inside a plateau of $\lceil H/C \rceil$ is dominated by the smallest C in that plateau. At every strict drop, no earlier point has as small a state coordinate and no later point has as small an alphabet. For $C > H$ the state coordinate is already B, so $C = H$ dominates all later points. Exact attainability holds for every C by the normalized phase-local constructions. □

At fixed C, the phase state premium is

$$\Delta^{\text{phase}}_Q(B,C) = B\left\lceil \frac{H(B)}{C} \right\rceil - S_{\text{raw}}(C) \geq 0.$$

For $C \geq 2$, equality holds exactly when

$$B = M, \qquad C \geq H(B).$$

Since p, q are coprime, $B = M$ gives

$$H(B) = \min(p,q) + 1.$$

Thus additional stack symbols can absorb the stronger protected phase information while retaining raw-optimal state count.

10.1 The audited raw-safe nesting

There is one important scope distinction. Model R forbids epsilon-normalization on encoded residual configurations. The phase-local optimum is raw and epsilon-free for $C \geq 2$, but the $C = 1$ optimum uses one forced bottom-cell normalization. Hence full $C \geq 1$ phase-local and raw classes are not literally nested.

Restrict to at least two work symbols. Let $\mathfrak{K}^{\geq 2}_{\text{raw}}$ be Model R with $C \geq 2$, and let $\mathfrak{K}^{\text{raw}}_{\text{phase}}(B)$ be the phase-local contract restricted to $C \geq 2$.

Theorem 10.3 (Exact nested raw-safe comparison)**.** *The restricted phase-local contract refines the restricted raw contract, and*

$$\operatorname{Pareto}(\mathfrak{K}^{\geq 2}_{\text{raw}}) = \{(M,3)\}.$$

The two restricted Pareto frontiers are equal exactly when

$$\boxed{B = M \quad \textit{and} \quad \min(p,q) = 1.}$$

In every other case the refinement is strict.

Proof. The $C \geq 2$ phase constructions are raw, so the contract inclusion is genuine. Every phase point has state at least $B \geq M$ and alphabet at least 3, hence $(M, 3)$ dominates. Reverse dominance requires a phase point at most $(M, 3)$, so $C = 2$, $B = M$, and $\lceil H/2 \rceil = 1$, equivalently $H = 2$. At $B = M$, coprimality gives $H = \min(p, q) + 1$, hence $\min(p, q) = 1$. $\square$

Theorem 10.4 (Direct full-frontier comparison)**.** *Over all $C \geq 1$,*

$$\mathrm{Pareto}(\mathfrak{K}_{\mathrm{raw}}) \preceq_P \mathrm{Pareto}(\mathfrak{K}_{\mathrm{phase}}(B)),$$

and the dominance is strict except exactly at

$$(p, q, B) = (1, 1, 1),$$

where both frontiers equal

$$\{(2, 2), (1, 3)\}.$$

Proof. For the $C = 1$ phase point,

$$BH(B) \geq 2B \geq 2M \geq p + q,$$

so the raw point $(p + q, 2)$ dominates it. For every phase point with $C \geq 2$, the raw point $(M, 3)$ dominates because its state coordinate is at least $B \geq M$. Reverse dominance forces a phase point below $(p + q, 2)$, necessarily the $C = 1$ point with $BH(B) \leq p + q$. Equality throughout the displayed chain occurs only at $(1, 1, 1)$. $\square$

11 What this theory does and does not claim

The theory is category-relative by design.

- It does not claim a representation-independent minimal DPDA.
- The raw frontier is exact for Model R, not for all recognizers of $L_{p,q}$.
- The phase staircase is exact for the normalized phase-local contract.
- Numerical resource penalties do not follow from observable refinement alone; a capacity or down-set witness is required for strictness.
- Terminal protected cores need not be unique. A theorem forcing all quotient descents to join would be a confluence problem, not part of the present minimality theorem.
- The unrestricted state–stack Pareto frontier and bounded-push hierarchy are separate problems.

12 Conclusion

Once the admissible architectures and protected physical information are stated explicitly, exact pushdown lifts admit a nontrivial reduction theory. SCQ quotients remove genuine coordinate redundancy, BBR equivalence removes harmless coding choices, protected interfaces prevent semantic trivialization, and Dickson finiteness turns recoding classes into finite Pareto objects. Contract refinement then has a monotonicity law, while protected-role capacity produces intrinsic strictness certificates.

The rational balance family shows that these abstract objects are computable. Stronger local visibility can move an entire state–stack frontier, yet enough stack alphabet can absorb the state premium. This is precisely the phenomenon that a single scalar “minimum number of states” cannot express.

A Typed protected contracts and BBR equivalence

For complete type correctness, a protected representation contract is written

$$\mathfrak{K} = (\mathfrak{C}, \{\lambda_{\mathcal{L}}\}_{\mathcal{L}\in\mathfrak{C}}),$$

where $\mathfrak{C}$ is a BBR-closed class of finite-coordinate exact sectioned lifts of one semantic system and each $\lambda_{\mathcal{L}} : C^{\mathrm{reach}}_{\mathcal{L}} \to \Lambda$ is a protected interface. The family is required to be compatible with BBR conjugacies: if $F : \mathcal{L} \to \mathcal{L}'$ is a BBR equivalence, then $\lambda_{\mathcal{L}} = \lambda_{\mathcal{L}'} \circ F$. In the body we suppress the object subscript when no ambiguity is possible.

Proposition A.1 (BBR is an equivalence relation)**.** *Bounded bi-recoding is reflexive, symmetric, and transitive on finite-coordinate protected lifts.*

Proof. The identity code gives reflexivity. The defining bounded non-erasing inverse on reachable stack languages gives symmetry. If block codes φ and ψ have maximum block lengths B_φ, B_ψ, then $\psi \circ \varphi$ has maximum block length at most $B_\varphi B_\psi$; the inverse is the reverse composition of the bounded inverses. The induced configuration bijections compose and continue to intertwine semantics, section, transitions, and the protected-interface family. Hence transitivity holds. □

Under contract refinement $\mathfrak{K}' \succeq \mathfrak{K}$, the representation class is included and the protected interfaces refine objectwise. Therefore the resource-set inclusion used in Theorem 7.4 is literal, rather than an informal comparison between unrelated coordinate systems.

B Endpoint resource theorems used in the rational-balance application

The rational-balance application combines previously established endpoint optima with the new protected-contract formalism. The logical inputs are:

1. in the broad deterministic pushdown/right-endmarker exact-lift class, the exact state minimum is $\max(p, q)$ and is attained with two work symbols plus the bottom marker;
2. in the canonical deferred-SEARCH literal-demand class, the exact state minimum is

$$D_{p,q} = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1,$$

again attainable with three total stack symbols;
3. in raw residual-faithful Model R, with C work symbols,

$$S_{\mathrm{raw}}(C) = \begin{cases} p+q, & C = 1, \\ \max(p,q), & C \geq 2; \end{cases}$$

4. in the normalized phase-local class,

$$Q^{\min}_{\mathrm{phase}}(B, C) = B\left\lceil \frac{H(B)}{C} \right\rceil, \qquad H(B) = \frac{p}{\gcd(B,p)} + \frac{q}{\gcd(B,q)}.$$

For reviewability, the proof interfaces are as follows. The deferred lower bound exposes the $2h+1$ residual debts $-hm, \ldots, 0, \ldots, hm$ at the same exhausted work stack; a common distinguishing continuation then forces distinct control states, while the centered rail attains the bound. The raw frontier uses positive- and negative-side synchronization to give $|Q| \geq \max(p, q)$ for every nonempty work alphabet, strengthens this to $p + q$ with one work symbol by cross-sign unary separation, and is attained by the one- and two-work-symbol residual encodings. The normalized phase-local formula counts, in each residue class modulo B, the $H(B)$ signed phase roles that must be exposed through C work symbols and then sums the sharp per-class bound; the matching construction is raw for $C \geq 2$. The broad unrestricted state theorem combines the positive- and negative-side phase-exposure bounds $|Q| \geq p$ and $|Q| \geq q$ with the explicit three-symbol residual realization having $\max(p, q)$ states.

These endpoint results are inputs, not re-labelled contributions of the present paper. They are included here at the level needed to audit every numerical comparison, while the new results are the SCQ/BBR reduction theory, protected-contract order, Pareto monotonicity/strictness criteria, and the synthesis of the endpoint optima into exact contract frontiers. The $C = 1$ phase optimum uses one forced bottom-cell normalization and is therefore compared directly with the full raw frontier rather than being treated as a literal subclass of raw Model R. For $C \geq 2$ the phase construction is raw, which is precisely the nested scope of Theorem 10.3.

Statements and Declarations

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